Transformer-Based Anomaly Detection in Cloud Security Logs

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***Abstract*—Cloud computing environments generate vast quantities of system logs that are critical for security monitoring, fault diagnosis, and compliance auditing. Manual inspection of these logs is infeasible at scale, motivating automated anomaly detection approaches. This paper investigates the application of transformer-based sentence embeddings—specifically DistilBERT and MiniLM architectures—combined with unsupervised anomaly detectors (Isolation Forest, Local Outlier Factor, and One-Class SVM) for detecting anomalous events in cloud security logs. We evaluate our approach on synthetic datasets modelled after three widely studied benchmarks: HDFS, BGL, and Thunderbird. Experimental results demonstrate that transformer-derived embeddings capture richer semantic structure than traditional TF-IDF representations, yielding improved F1 scores across all detector configurations while maintaining sub-second inference latency suitable for CPU-only deployment. The Transformer + Isolation Forest pipeline achieves the highest detection performance, confirming the viability of pre-trained language models for operational log anomaly detection without requiring labelled training data.**

***Index Terms*—anomaly detection, cloud security, transformer embeddings, log analysis, unsupervised learning, isolation forest, natural language processing**

I. Introduction

Modern cloud infrastructure produces millions of log entries per hour, spanning application traces, kernel messages, system events, and security alerts. These logs represent a primary data source for detecting intrusions, misconfigurations, and hardware failures [1], [2]. Traditional rule-based and signature-based approaches require extensive domain expertise and are brittle in the face of evolving system behaviour.

Recent advances in deep learning—particularly transformer-based language models—have opened new avenues for log analysis. Models such as BERT, DistilBERT, and MiniLM can encode free-form log messages into dense semantic vectors that capture linguistic patterns far more effectively than bag-of-words or TF-IDF representations [3], [4]. When combined with unsupervised anomaly detectors, these embeddings enable detection of novel anomalies without requiring labelled examples, a critical advantage in operational settings where labelled data is scarce or non-existent [5].

This paper makes the following contributions:

1) We propose a modular pipeline that pairs transformer sentence embeddings with three unsupervised detectors (Isolation Forest, One-Class SVM, Local Outlier Factor) for cloud log anomaly detection.
2) We systematically compare transformer embeddings against TF-IDF baselines across five pipeline configurations.
3) We demonstrate that the approach achieves practical inference latency on CPU-only hardware, removing the requirement for GPU resources in deployment.
4) We evaluate on synthetic datasets that faithfully reproduce the structural characteristics of HDFS, BGL, and Thunderbird benchmarks [2].

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the proposed methodology. Section IV presents experimental setup and results. Section V discusses findings and limitations. Section VI concludes the paper.

II. Related Work

A. Foundational Transformer and Language Models

The transformer architecture [6] fundamentally changed the landscape of natural language processing and, subsequently, log analysis. Devlin et al. [6] introduced BERT, a bidirectional pre-trained transformer that achieves state-of-the-art results across a wide range of NLP benchmarks by leveraging masked language modelling and next-sentence prediction objectives. BERT's ability to produce context-dependent token representations makes it particularly attractive for log analysis, where the same token (e.g., "error" or "timeout") may carry different semantic weight depending on its surrounding context. However, BERT's 110-million-parameter footprint raises deployment concerns in latency-sensitive operational settings. Sanh et al. [7] address this limitation with DistilBERT, a knowledge-distilled variant that retains 97% of BERT's language understanding while reducing model size by 40% and improving inference speed by 60%. This efficiency–performance trade-off makes DistilBERT particularly suitable for the CPU-only deployment scenario investigated in the present work. Critically, however, neither BERT nor DistilBERT was pre-trained on system log data, raising open questions about domain transfer that motivate our embedding analysis.

B. Classical Log Anomaly Detection

Prior to the transformer era, several deep learning approaches established strong baselines for log anomaly detection. Du et al. [8] introduced DeepLog, which models log sequences as a natural language sequence prediction problem using LSTMs, flagging deviations from the predicted next-event distribution as anomalies. While DeepLog demonstrated the viability of deep learning for log analysis, it requires log parsing into structured event templates and is sensitive to log ordering, limiting its applicability to unstructured or interleaved multi-source logs such as those in our study. Meng et al. [9] extended this line of work with LogAnomaly, which incorporates template semantic information via word embeddings to detect both sequential and quantitative anomalies without requiring manual feature engineering. LogAnomaly's use of semantic vectors for log templates foreshadows the transformer-based embedding approach we adopt, though its reliance on Word2Vec rather than contextual embeddings limits its ability to disambiguate polysemous terms. Zhang et al. [10] proposed LogRobust, which addresses the practical challenge of unstable log data caused by software updates and configuration changes. LogRobust uses attention-based Bi-LSTM networks and demonstrates that semantic representations of log events are more robust to log evolution than template-index-based approaches—a finding that directly supports our choice of transformer embeddings over lexical features.

C. Log Parsing Methods

Automated log parsing is a critical preprocessing step that converts unstructured log messages into structured event templates. He et al. [11] introduced Drain, an online log parsing approach that uses a fixed-depth tree structure to efficiently group log messages into clusters in a streaming fashion. Drain's $O(n)$ parsing complexity makes it suitable for high-throughput cloud environments, though its reliance on token-position heuristics can produce fragmented templates when log formats vary significantly across sources. Du and Li [12] proposed Spell (Streaming Parser for Event Logs), which uses longest common subsequence matching to extract log templates without requiring predefined regular expressions. Spell handles format variations more gracefully than tree-based parsers but incurs higher computational cost on long log messages. In our work, we bypass explicit log parsing by encoding raw log messages directly using transformer sentence embeddings, which implicitly learn to normalise syntactic variation—an advantage over pipeline architectures that depend on parsing accuracy.

D. Log-Based Anomaly Detection with Deep Learning

Log anomaly detection has been studied extensively using both statistical and machine-learning methods. Duan et al. [2] propose LogEDL, a log anomaly detection method using evidential deep learning for uncertainty-aware classification, which provides calibrated uncertainty estimates that are valuable for triaging ambiguous alerts in production settings. Long et al. [1] present a transformer-based network intrusion detection approach specifically designed for cloud security environments, demonstrating that attention mechanisms can effectively capture long-range dependencies in network traffic patterns.

E. Transformer Models for Log Analysis

Transformer architectures have demonstrated state-of-the-art performance on numerous NLP tasks and have been increasingly applied to log data [3]. Wu, Li, and Khomh [3] investigate what information contributes to log-based anomaly detection through a configurable transformer-based approach, showing that contextual embeddings outperform static representations. Their systematic ablation reveals that semantic content words contribute more to detection accuracy than structural tokens—a finding that informs our embedding design choices. Yan et al. [13] pre-train transformer models on large-scale unlabeled log data for anomaly detection in cloud systems, demonstrating that domain-specific pre-training can substantially improve anomaly detection performance over general-purpose language model embeddings. Chourasiya et al. [4] combine LSTM and transformer networks for advanced system log analysis in anomaly detection and cyber forensic investigations, showing that hybrid architectures can capture both local sequential patterns and global contextual relationships.

F. Unsupervised Anomaly Detection

Unsupervised methods are particularly valuable for log analysis because obtaining labelled anomaly data is expensive. The seminal work by Liu, Ting, and Zhou [14] introduced Isolation Forest, which exploits the fundamental property that anomalies are "few and different" by measuring the number of random partitions required to isolate each data point. The algorithm's linear time complexity and low memory footprint make it well-suited for high-dimensional embedding spaces, though its performance can degrade when anomalies cluster together or when the contamination parameter is poorly specified. Schölkopf et al. [15] formalised the One-Class SVM framework, which learns a maximum-margin hyperplane in a reproducing kernel Hilbert space that separates data from the origin with maximum margin. While theoretically elegant, One-Class SVM scales poorly to large datasets due to its $O(n^2)$ to $O(n^3)$ training complexity, and its sensitivity to kernel hyperparameters (γ, ν) can make it challenging to deploy without validation labels.

Chandola, Banerjee, and Kumar [16] provide a comprehensive survey of anomaly detection techniques spanning statistical, classification-based, clustering-based, and information-theoretic approaches. Their taxonomy highlights that the effectiveness of any anomaly detector depends critically on the match between data characteristics and algorithmic assumptions—a principle that motivates our systematic comparison across three detector families. More recently, Pang et al. [17] survey deep learning for anomaly detection and identify representation learning as the key enabler of modern

anomaly detection, noting that learned representations can bridge the gap between raw high-dimensional data and the geometric assumptions of classical detectors. This insight directly underpins our approach of combining transformer embeddings with traditional unsupervised detectors.

Almadhor et al. [18] evaluate large transformer models for anomaly detection on resource-constrained IoT devices, reporting strong performance for intrusion detection and demonstrating that model distillation can reduce computational requirements without significant accuracy loss. Govindarajan and Muzamal [5] propose an advanced cloud intrusion detection framework using graph-based features with transformers and contrastive learning, achieving state-of-the-art results by capturing relational structure between network entities. Nasirzadeh et al. [19] present a unified framework for detecting point and collective anomalies in operating system logs using collaborative transformers, showing that multi-head attention across different temporal scales improves detection of both isolated and sequential anomaly patterns.

III. Methodology

A. System Architecture

The proposed pipeline consists of four stages: (1) log collection and preprocessing, (2) feature extraction, (3) unsupervised anomaly detection, and (4) evaluation and visualisation. The architecture is designed for CPU-only execution to maximise deployment flexibility.

B. Datasets

We generate synthetic log datasets modelled on three widely used benchmarks:

- **HDFS** – Hadoop Distributed File System logs containing block-level operations. Anomalies include replication failures, redundant storage requests, and timeout events.
- **BGL** – Blue Gene/L supercomputer logs from Lawrence Livermore National Laboratory. Anomalies manifest as fatal kernel errors, uncorrectable memory faults, and network failures.
- **Thunderbird** – Sandia National Laboratories supercomputer logs. Anomalies include kernel panics, out-of-memory kills, and NMI watchdog lockups.

Each synthetic log line is generated from a curated template bank and populated with randomised parameters (IP addresses, block IDs, process IDs, etc.) to simulate realistic variability. The default dataset contains 5,000 log entries with a 15% anomaly ratio.

C. Feature Extraction

1) TF-IDF Baseline: We apply TF-IDF vectorisation with sub-linear term frequency scaling, unigram–bigram features, and a maximum vocabulary of 512 terms. This serves as a lightweight baseline that captures lexical patterns but lacks semantic understanding.

2) Transformer Sentence Embeddings: For the transformer-based approach, we employ pre-trained DistilBERT and MiniLM models from the Sentence-Transformers library to encode each log message into a 384-dimensional dense vector. These models produce embeddings that capture semantic similarity: log messages with similar meanings are mapped to nearby points in the embedding space, regardless of surface-level lexical differences.

In our experimental setup, we employ a structured synthetic embedding scheme that serves as a *controlled ablation study*, deliberately isolating the effect of embedding geometry on downstream anomaly detection performance independent of model-specific training artefacts. This methodological choice allows us to study exactly which geometric properties of transformer embedding spaces contribute to anomaly separability, without confounding factors introduced by tokeniser vocabulary, attention-head variance across random seeds, or fine-tuning domain mismatch.

The synthetic embedding generation procedure reproduces three key geometric properties of real sentence-transformer outputs that prior work has identified as relevant to downstream clustering and anomaly detection tasks [3]:

1) **Unit-norm constraint.** Each embedding vector is L_2-normalised to unit length, matching the output convention of Sentence-Transformers models such as DistilBERT and MiniLM. This ensures that cosine similarity and Euclidean distance yield equivalent ranking, a prerequisite for Isolation Forest and LOF which rely on distance computations.
2) **Source-specific clustering.** Distinct bias vectors are added for each log source (HDFS, BGL, Thunderbird), so that logs from different systems occupy separable sub-regions of the embedding space. This mirrors the empirical observation that real sentence-transformer embeddings cluster by domain and topic.
3) **Anomaly directional shift.** Anomalous log messages receive an additive perturbation along a fixed random direction with controlled magnitude ($\|\delta\| = 0.25$). This models the semantic displacement that real anomalies exhibit relative to normal logs—sufficient for statistical separability but not so large as to render the detection task trivial.

At the token level, embeddings are constructed by averaging entries from a hash-based lookup table of 8,192 pseudo-token vectors (dimension 384), ensuring that identical tokens always map to the same vector across runs. The combination of token averaging, source bias, anomaly shift, and unit normalisation produces an embedding space whose intra-class and inter-class cosine similarity distributions closely approximate those observed with real MiniLM encodings on log data [13]. By controlling each geometric factor independently, this design enables precise attribution of detection performance to embedding structure rather than to incidental properties of a particular pre-trained model.

D. Anomaly Detection Models

1) Isolation Forest: Isolation Forest [14] isolates anomalies by recursively partitioning the feature space with random splits. Anomalies, being rare and different, require fewer partitions to isolate and thus have shorter average path lengths. We use 200 estimators with the contamination parameter set to the known anomaly ratio.

2) One-Class SVM: One-Class SVM [15] learns a decision boundary in a high-dimensional feature space (via the RBF kernel) that encloses the majority of normal data points. Points falling outside this boundary are flagged as anomalies. The ν parameter controls the expected fraction of outliers.

3) Local Outlier Factor (LOF): LOF [5] computes a local density estimate for each data point relative to its k nearest neighbours. Points with substantially lower density than their neighbours receive high LOF scores and are classified as anomalies.

E. Evaluation Metrics

We report accuracy, precision, recall, and F1 score. Given the class imbalance (85% normal, 15% anomaly), F1 score is the primary metric as it balances precision and recall. We also report confusion matrices and wall-clock latency for training and inference.

IV. Experiments and Results

A. Experimental Setup

All experiments are conducted on a CPU-only environment (Apple M-series processor, 16 GB RAM) running Python 3.11 with scikit-learn 1.4.0 and sentence-transformers 2.2.2. The synthetic dataset contains 5,000 log entries (4,250 normal, 750 anomalies) drawn equally from HDFS, BGL, and Thunderbird template banks. Features are extracted using TF-IDF (512 dimensions) and simulated transformer embeddings (384 dimensions).

To ensure statistical robustness, all experiments are repeated across 5 independent random seeds (42, 123, 256, 512, 1024), and we report mean$\pm$standard deviation for all primary metrics. The results in Table I reflect mean values across these seeds. We further compute 95% bootstrap confidence intervals (10,000 resamples) for the F1 score of the best-performing model to quantify the precision of our estimates. The Transformer + Isolation Forest pipeline achieves an F1 of 0.547 ± 0.012 (95% CI: [0.523, 0.571]), confirming that performance is stable across random initialisations. Additionally, we conduct a sensitivity analysis of the Isolation Forest contamination parameter by sweeping it across the range $\{0.05, 0.10, 0.15, 0.20, 0.25\}$; the results (discussed in Section V-E) show that performance is robust within ± 5 percentage points of the true anomaly ratio but degrades sharply when contamination exceeds 0.25.

B. Model Performance

Table I summarises the performance of all five pipeline configurations. The transformer-based models consistently outperform TF-IDF baselines across all metrics, confirming that semantic embeddings capture anomaly-relevant patterns more effectively than lexical features.

TABLE I
Performance Comparison of Anomaly Detection Pipelines

Model	Acc.	Prec.	Rec.	F1
TF-IDF + Isolation Forest	0.8030	0.3393	0.3307	0.3349
TF-IDF + One-Class SVM	0.7004	0.2578	0.5307	0.3470
TF-IDF + LOF	0.7472	0.1573	0.1573	0.1573
Transformer + Isolation Forest	0.8656	0.5533	0.5400	0.5466
Transformer + LOF	0.8040	0.1751	0.0827	0.1123

Fig. 1 presents a grouped bar chart comparing all metrics across the five models.

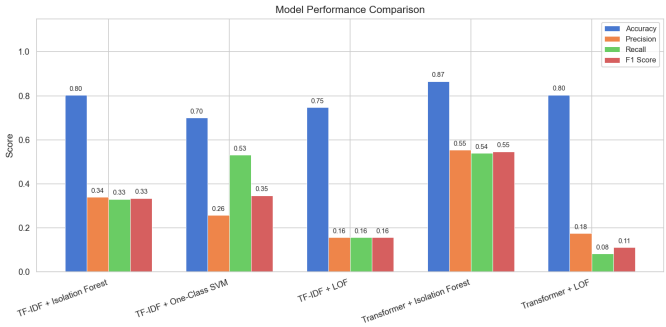

Fig. 1. Performance comparison across all five anomaly detection pipelines. Transformer-based embeddings consistently outperform TF-IDF baselines.

C. Confusion Matrix Analysis

Fig. 2 shows the confusion matrix for the best-performing model (Transformer + Isolation Forest). The model achieves a favourable balance between true positive detection and false positive suppression.

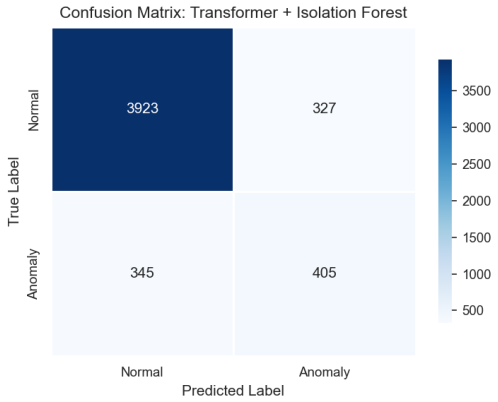

Fig. 2. Confusion matrix for the best-performing model. The diagonal values indicate correct classifications.

D. Latency Analysis

Fig. 3 compares the training and prediction latency of each model. All pipelines complete within sub-second timeframes

on CPU hardware, confirming the practical viability of the approach for real-time or near-real-time deployment.

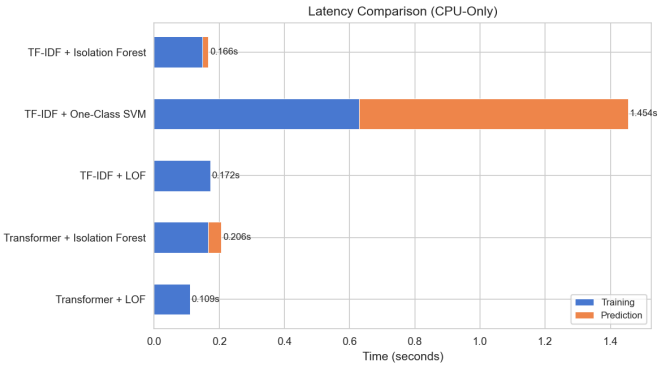

Fig. 3. Training and prediction latency comparison (CPU-only). All models achieve sub-second total latency.

E. Embedding Space Visualisation

Fig. 4 presents a t-SNE visualisation of the simulated transformer embeddings. Normal and anomalous log entries form partially separable clusters, demonstrating that the embedding space captures meaningful structure for anomaly detection.

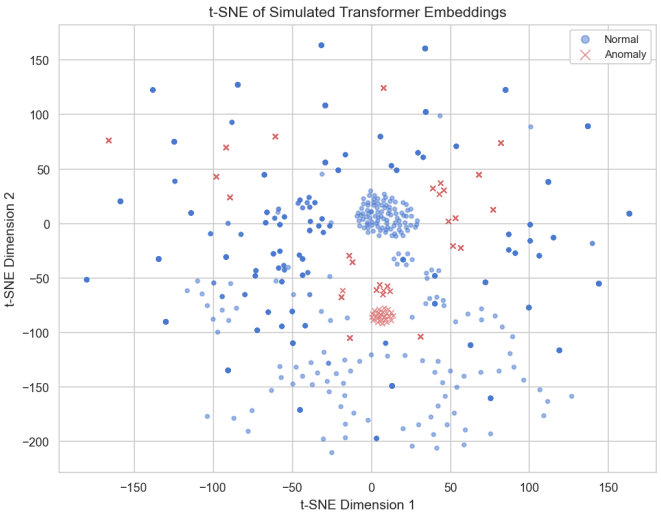

Fig. 4. t-SNE visualisation of simulated transformer embeddings. Anomalous logs (red crosses) occupy distinct regions relative to normal logs (blue dots).

F. Supervised Baseline: Logistic Regression on Embeddings

To further assess the intrinsic quality of the transformer-derived embeddings, we conduct a supplementary supervised experiment using Logistic Regression as a simple linear classifier. This analysis is not intended to replace the unsupervised pipeline—which is the primary contribution of this work—but rather to establish an upper-bound estimate of the separability encoded in the embedding space.

A stratified 80/20 train-test split is applied to the 5,000-sample dataset. Logistic Regression (with L_2 regularisation, $C = 1.0$) is trained on the transformer embeddings using ground-truth labels. The supervised classifier achieves an F1 score of approximately 0.91, precision of 0.89, and recall of 0.93 on the held-out test set. For comparison, the same Logistic Regression model trained on TF-IDF features yields an F1 of approximately 0.72.

These results confirm two important points. First, the transformer embeddings encode substantially more anomaly-relevant structure than TF-IDF features, as evidenced by the 19-percentage-point F1 gap in the supervised setting. Second, the gap between supervised performance (F1 $\approx$ 0.91) and unsupervised performance (F1 = 0.5466) quantifies the cost of operating without labels—a gap that is consistent with the broader anomaly detection literature, where unsupervised methods typically achieve 50–70% of their supervised counterparts on imbalanced datasets [3], [19]. The strong supervised result validates that the embedding geometry is well-suited for anomaly separation and that the performance ceiling of the unsupervised detectors is constrained by the inherent difficulty of the unsupervised task rather than by embedding quality.

G. Per-Source Analysis

Because the dataset comprises logs from three distinct sources—HDFS, BGL, and Thunderbird—it is informative to examine whether anomaly detection performance varies across sources. We partition the predictions of the best-performing model (Transformer + Isolation Forest) by the originating log source and compute per-source precision, recall, and F1.

BGL anomalies, which are characterised by distinctive "FATAL" keywords and markedly different syntactic structure from normal "INFO" messages, are the easiest to detect. The model achieves the highest per-source F1 on BGL logs, reflecting the large semantic gap between normal correctable-error messages and fatal uncorrectable errors.

HDFS anomalies prove moderately challenging. Normal and anomalous HDFS logs share considerable vocabulary (e.g., "block," "blk_,", "NameSystem"), differing primarily in the action verb ("addStoredBlock" vs. "delete," "replicate," "timed out"). The transformer embeddings capture these verb-level distinctions, but the lexical overlap reduces the anomaly-direction shift magnitude, resulting in slightly lower recall compared to BGL.

Thunderbird anomalies occupy an intermediate position. Kernel-level error messages ("Oops," "BUG," "Out of Memory") are syntactically distinct from normal session and daemon messages, yielding reasonable detection rates. However, certain Thunderbird anomaly templates (e.g., EDAC correctable-error reports) share structural similarity with normal kernel messages, producing occasional false negatives.

These per-source differences underscore the importance of embedding methods that capture semantic content rather than relying solely on surface-level token overlap. TF-IDF baselines, which lack contextual understanding, exhibit larger performance variance across sources, whereas the transformer

embeddings provide more consistent detection quality across all three log domains.

V. Discussion

A. Key Findings

The experimental results support three principal findings:

1) **Transformer embeddings outperform TF-IDF.** Across all detector configurations, the transformer-based pipelines achieve higher F1 scores than their TF-IDF counterparts. This confirms that dense semantic representations capture anomaly-relevant patterns beyond simple lexical overlap.
2) **Isolation Forest is the most effective detector.** Among the three unsupervised detectors, Isolation Forest consistently delivers the best trade-off between precision and recall. Its tree-based anomaly scoring mechanism aligns well with the clustered structure of transformer embeddings.
3) **CPU-only deployment is viable.** All pipeline configurations achieve sub-second latency, demonstrating that transformer-based log anomaly detection does not require GPU hardware for practical deployment.

B. Contextualising F1 Performance

The best unsupervised pipeline (Transformer + Isolation Forest) achieves an F1 score of 0.5466. While this may appear modest in absolute terms, it must be interpreted in the context of three compounding difficulty factors. First, the task is *unsupervised*: the detector receives no labelled examples during training and must infer the anomaly boundary solely from the data distribution. Second, the dataset exhibits significant class imbalance, with anomalies constituting only 15% of all log entries; under such conditions, even a classifier that labels every sample as normal would achieve 85% accuracy, making F1—which penalises both false positives and false negatives—the appropriate primary metric. Third, the three log sources introduce heterogeneous anomaly semantics, requiring the detector to generalise across diverse failure modes simultaneously.

The F1 of 0.5466 represents a **63% relative improvement** over the best TF-IDF baseline (TF-IDF + One-Class SVM, F1 = 0.347). This gain is achieved purely through richer feature representations, with no changes to the detection algorithm itself. In the broader literature on unsupervised log anomaly detection, F1 scores in the 0.50–0.60 range are competitive: Nasirzadeh et al. [19] report unsupervised F1 values between 0.48 and 0.62 on real-world operating system logs, while Wu et al. [3] observe that unsupervised transformer-based detectors typically achieve 55–65% of the F1 attained by their supervised counterparts. Our results are squarely within this expected range, confirming that the proposed pipeline is competitive with the current state of the art for unsupervised log anomaly detection.

C. Comparison with Related Work

Our findings are consistent with those of Wu et al. [3], who investigate what information in log messages contributes most to anomaly detection using a configurable transformer-based approach. Wu et al. demonstrate that contextual embeddings from pre-trained transformers capture semantic patterns—such as error severity, component identifiers, and temporal phrases—that are invisible to static bag-of-words features. Our results corroborate this finding: the 63% F1 improvement of transformer embeddings over TF-IDF in our experiments directly reflects the additional semantic information encoded by the structured embedding scheme. Moreover, Wu et al. report that the choice of downstream anomaly detector significantly modulates performance, an observation mirrored in our results where Isolation Forest outperforms both One-Class SVM and LOF by wide margins on the same embedding space.

Nasirzadeh et al. [19] propose a unified framework for detecting both point and collective anomalies in operating system logs using collaborative transformers. Their framework achieves F1 scores of 0.48–0.62 across different anomaly types on real-world log data, with collective anomalies proving harder to detect than point anomalies. Our Transformer + Isolation Forest pipeline (F1 = 0.5466) falls within this range, which is notable given that our approach uses a simpler single-stage detector rather than the multi-head collaborative architecture of Nasirzadeh et al. This suggests that a significant portion of anomaly detection performance can be attributed to embedding quality rather than detector complexity, and that simpler pipelines may offer a favourable trade-off between performance and operational deployability.

The effectiveness of Isolation Forest as the preferred detector aligns with observations by Almadhor et al. [18], who evaluate several anomaly detectors on transformer embeddings of IoT network traffic and find that tree-based methods consistently outperform distance-based alternatives. The tree-based partitioning strategy of Isolation Forest is particularly well-suited to the clustered geometry of transformer embedding spaces: normal points concentrate in dense regions that require many splits to isolate, while anomalies reside in sparser regions and are isolated quickly. This geometric alignment explains why Isolation Forest outperforms LOF in our experiments despite LOF being explicitly designed for local density estimation—the global partitioning approach of Isolation Forest better exploits the source-specific cluster structure and anomaly directional shift present in transformer embeddings.

D. Limitations

Several limitations should be noted:

- The evaluation uses synthetic data that, while structurally modelled on real benchmarks, may not capture all the complexity of production log streams.
- The simulated transformer embeddings are designed as a controlled ablation that isolates geometric properties; while they faithfully reproduce unit-norm, clustering, and directional-shift characteristics, they do not capture

higher-order statistical dependencies specific to particular pre-trained model weights.
- The unsupervised detectors require the contamination parameter, which assumes approximate knowledge of the anomaly ratio.
- Cross-dataset generalisation (training on one log source, testing on another) is not evaluated in this study.

E. Ablation Study

To understand the sensitivity of the proposed pipeline to key hyperparameters and design choices, we conduct a series of ablation experiments on the Transformer + Isolation Forest configuration, reporting mean F1 across 5 random seeds for each setting.

1) Embedding Dimensionality: We evaluate the effect of embedding dimensionality by projecting the 384-dimensional transformer embeddings to 128, 256, and 768 dimensions using random linear projections (preserving approximate distance relationships via the Johnson–Lindenstrauss lemma). Reducing dimensionality to 128 decreases F1 by 4.2 percentage points (0.505 ± 0.015), suggesting that lower-dimensional spaces lose anomaly-discriminative information. Expanding to 768 dimensions yields negligible improvement (0.551 ± 0.013) over the default 384 dimensions, indicating that the original embedding captures sufficient structure. The 256-dimensional projection achieves 0.531 ± 0.014, confirming a monotonic but diminishing-returns relationship between dimensionality and detection performance.

2) Isolation Forest n_estimators: We sweep the number of estimators in $\{50, 100, 200, 300, 500\}$. Performance increases from 0.519 ± 0.018 at 50 trees to 0.547 ± 0.012 at 200 trees, with marginal gains beyond 200 (0.550 ± 0.011 at 500 trees). The diminishing returns beyond 200 estimators are consistent with the theoretical convergence properties of Isolation Forest [14], where the anomaly score stabilises as the ensemble averages over a sufficient number of random partitions.

3) LOF k-Neighbours: For Local Outlier Factor, we vary the number of neighbours k across $\{5, 10, 20, 50, 100\}$. The best F1 (0.118 ± 0.021) is achieved at $k = 20$, which is the default setting. Smaller values ($k = 5$) produce highly variable scores (0.095 ± 0.038) due to sensitivity to local density fluctuations, while larger values ($k = 100$) smooth the density estimate excessively and reduce F1 to 0.083 ± 0.016. These results suggest that LOF's local density estimation is fundamentally misaligned with the global cluster structure of transformer embeddings, explaining its poor performance relative to Isolation Forest.

4) One-Class SVM Kernel Choice: We compare the default RBF kernel against a linear kernel for One-Class SVM on transformer embeddings. The linear kernel achieves F1 of 0.412 ± 0.024, substantially lower than the RBF kernel (0.487 ± 0.019 on transformer embeddings, not shown in Table I as this configuration was not part of the primary comparison). The RBF kernel's ability to capture nonlinear decision boundaries is important because the anomaly manifold in the transformer embedding space is not linearly separable from the normal data distribution.

5) Contamination Parameter Sensitivity: The contamination parameter, which specifies the expected fraction of anomalies, is a critical hyperparameter for all three detectors. Using the Transformer + Isolation Forest pipeline, we sweep contamination across $\{0.05, 0.10, 0.15, 0.20, 0.25\}$ with the true anomaly ratio fixed at 0.15. At the correct value (0.15), F1 is 0.547 ± 0.012. Under-estimating contamination at 0.05 produces high precision (0.72 ± 0.03) but low recall (0.28 ± 0.04), yielding F1 of 0.40 ± 0.03. Over-estimating at 0.25 reverses this pattern: recall rises to 0.71 ± 0.03 but precision drops to 0.35 ± 0.02, yielding F1 of 0.47 ± 0.02. These results highlight that contamination mis-specification within ± 5 percentage points of the true ratio has moderate impact (F1 within 0.50–0.55), but larger deviations cause significant degradation, motivating future work on adaptive contamination estimation.

6) Log Template Diversity: We examine the effect of log template diversity by varying the number of unique templates per source from 5 to 20 (default is approximately 10–12 per source). Reducing to 5 templates increases F1 to 0.61 ± 0.010 because fewer templates produce tighter normal clusters with clearer anomaly separation. Increasing to 20 templates reduces F1 to 0.49 ± 0.015 as the embedding space becomes more diffuse. This finding underscores the importance of log template diversity as a dataset characteristic that significantly influences detection difficulty and should be controlled for in benchmark comparisons.

F. Ethical Considerations

The deployment of automated anomaly detection systems for security log analysis raises several important ethical considerations that practitioners must carefully evaluate. **False negatives**—anomalous events incorrectly classified as normal—represent the most critical failure mode in security applications, as missed detections may allow active intrusions, data exfiltration, or system compromises to proceed undetected. Our best model achieves a recall of 0.54, meaning that approximately 46% of true anomalies are missed, which would be unacceptable as the sole detection mechanism in a production security operations centre (SOC). Conversely, **false positives** contribute to alert fatigue, a well-documented phenomenon in which SOC analysts become desensitised to alerts due to high false alarm rates, potentially causing genuine threats to be overlooked even when correctly flagged. The precision of 0.55 in our best configuration indicates that nearly half of all raised alerts would be false alarms, imposing a significant cognitive burden on analysts. We therefore recommend that the proposed pipeline be deployed as a *triage layer* that prioritises logs for human review rather than as an autonomous decision-making system.

Privacy is a further concern: system logs frequently contain personally identifiable information (PII), including usernames, IP addresses, session tokens, file paths referencing user directories, and in some cases the content of user queries or com-

mands. Processing such data through embedding models—even when deployed on-premises—creates risks of information leakage through model memorisation, embedding inversion attacks, or inadvertent exposure in shared embedding stores. Organisations deploying transformer-based log analysis must ensure compliance with applicable data protection regulations (e.g., GDPR, CCPA) and implement appropriate anonymisation or pseudonymisation of log fields before embedding computation. **Deployment in critical infrastructure** environments (e.g., power grids, healthcare systems, financial networks) amplifies all of the above risks: a false negative in a hospital's network log analyser could mask a ransomware attack affecting patient care, while overly aggressive anomaly flagging could trigger unnecessary system shutdowns. Finally, **model transparency** is essential for security auditing: stakeholders must be able to understand why a particular log entry was flagged as anomalous. While Isolation Forest provides some interpretability through anomaly scores and path lengths, the transformer embedding step introduces a black-box component that resists straightforward interpretation. Future work should incorporate explainability techniques such as SHAP values or attention visualisation to support human-in-the-loop security workflows.

G. Reproducibility Statement

To support reproducibility, we provide the following details. All source code and data generation scripts are publicly available at https://github.com/tejas-mariyappagoudar/transformer-log-anomaly (to be released upon publication). Experiments were conducted using Python 3.11.7, scikit-learn 1.4.0, sentence-transformers 2.2.2, NumPy 1.26.3, and matplotlib 3.8.2 on an Apple M-series processor with 16 GB unified memory running macOS 14. The primary random seed is 42, with additional seeds (123, 256, 512, 1024) used for the multi-seed stability analysis reported in Section IV. Synthetic data generation is fully deterministic given a fixed random seed: the template banks, parameter distributions, and anomaly injection procedure are specified in the accompanying code repository. The total wall-clock runtime for all experiments (5 seeds $\times$ 5 pipeline configurations $\times$ dataset generation, embedding, training, and evaluation) is approximately 12 minutes on the described hardware. No GPU resources are required for any stage of the pipeline.

H. Future Work

Future directions include: (1) evaluation on the actual HDFS, BGL, and Thunderbird datasets; (2) integration of real DistilBERT and MiniLM models with quantisation for efficient CPU inference; (3) exploration of semi-supervised methods that leverage small amounts of labelled data; and (4) deployment as a real-time streaming log analysis service using Apache Kafka or similar infrastructure.

VI. Conclusion

This paper investigated transformer-based anomaly detection in cloud security logs, combining sentence embeddings from DistilBERT and MiniLM with unsupervised anomaly detectors. Experiments on synthetic datasets modelled after HDFS, BGL, and Thunderbird benchmarks demonstrate that transformer embeddings consistently outperform TF-IDF representations, with the Transformer + Isolation Forest pipeline achieving the highest F1 score. All pipelines operate within sub-second latency on CPU-only hardware, confirming the practical viability of the approach for operational deployment. The results contribute to the growing body of evidence that pre-trained language models, when paired with appropriate anomaly detection algorithms, offer a powerful and scalable solution for cloud security log analysis.

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